Q11 — Deeper projects: Why do models hallucinate, and can they know when they are unsure?

DLS 2026 — Graduate AI-Internals Module

Framing the question

A language model trained by maximum-likelihood on next-token prediction is optimized to produce *fluent and plausible* continuations, not *truthful* ones. Hallucination—confident generation of content that is unsupported or false—is therefore not an incidental bug but a predictable consequence of the training objective, the data distribution, and the decoding procedure. Recent analyses argue that hallucinations arise partly because pretraining rewards guessing over abstention, and that standard benchmarks and reward schemes penalize "I don't know" responses, incentivizing confident fabrication [5, 9]. A closely related question is *calibration*: can a model's own probabilities, hidden states, or verbalized confidence reliably signal when it is likely to be wrong? Evidence is mixed—base models can be well calibrated on some tasks [4], but RLHF can degrade calibration [10], and factual knowledge appears to be linearly encoded in intermediate activations in ways that sometimes diverge from the model's own outputs [1, 2].

The projects below ask you to measure these phenomena carefully, probe their mechanistic origins, and build interventions. Each project lists concrete deliverables. Use open-weight models (e.g. Llama, Mistral, Pythia) so that internal states are accessible; proprietary APIs may be used for comparison.

1 Project A — Measuring and decomposing hallucination

Goal. Build a reproducible pipeline that elicits, detects, and categorizes hallucinations, and that separates *knowledge gaps* (the model never learned the fact) from *retrieval/decoding failures* (the model "knows" but outputs something false).

Tasks.

1. Assemble an evaluation set combining a closed-book factual QA benchmark, a summarization faithfulness benchmark, and an adversarial truthfulness benchmark [7]. Annotate each item with a ground-truth answer and supporting evidence.

2. Implement at least two automatic detectors: (i) an entailment/NLI-based faithfulness check against source documents [9], and (ii) a sampling-based consistency score such as SelfCheckGPT, which flags claims that vary across stochastic resamples [8].

3. Distinguish knowledge gaps from recall failures: for each item, probe whether the correct answer appears among top-k sampled completions or under constrained decoding even when greedy decoding is wrong.

Deliverables.

- A documented code repository and a dataset card with annotation guidelines and inter-annotator agreement on a 100-item subset.
- A report quantifying hallucination rate per task and per model size, with a confusion-style breakdown into knowledge-gap vs. recall-failure categories.
- An error taxonomy with at least 20 qualitative examples per category.

2 Project B — Confidence, calibration, and abstention

Goal. Determine how well a model "knows when it is unsure," comparing sequence likelihoods, verbalized confidence, and a trained "P(True)" head.

Tasks.

1. Compute reliability diagrams and expected calibration error (ECE) for (i) token/sequence probabilities, (ii) verbalized numeric confidence elicited by prompting, and (iii) a self-evaluation probe asking the model whether its own answer is correct [4].
2. Compare a base/instruct checkpoint against an RLHF-tuned variant to test whether alignment tuning degrades calibration [10].
3. Build an abstention policy: choose a confidence threshold that optimizes a scoring rule which rewards correct answers, mildly rewards abstention, and heavily penalizes confident errors, as advocated for discouraging guessing [5]. Plot accuracy-vs-coverage (selective prediction) curves.

Deliverables.

- Reliability diagrams and ECE/AUROC tables for all confidence signals across ≥ 2 models.
- A selective-prediction (risk–coverage) analysis with the derived abstention thresholds.
- A short discussion of which confidence signal is most trustworthy and under what distribution shifts it fails.

3 Project C — Probing internal "truthfulness" representations

Goal. Investigate whether a model's hidden states encode the truth of a statement more reliably than its output, and whether such representations can be read out without labels.

Tasks.

1. Train supervised linear probes on frozen activations to classify statements as true/false, sweeping over layers and token positions [1].
2. Reproduce an unsupervised method that finds a truth direction via logical-consistency constraints (Contrast-Consistent Search) and compare it to the supervised probe [2].
3. Measure the gap between what the probe "believes" and what the model *says*: identify cases where the internal probe is correct while the generated answer is a hallucination, and quantify how often this occurs.

Deliverables.

- Layerwise probe-accuracy curves plus a comparison of supervised vs. unsupervised probes.
- A quantitative "internal-knows-but-outputs-wrong" statistic with examples.
- A critical appendix on probing pitfalls (spurious features, probe capacity, generalization across datasets).

4 Project D — Interventions: grounding and decoding

Goal. Reduce hallucination via retrieval grounding and inference-time decoding changes, and measure the trade-off against fluency and task accuracy.

Tasks.

1. Implement a retrieval-augmented pipeline and measure how grounding affects faithfulness and knowledge-gap errors [6].
2. Implement a contrastive decoding intervention that amplifies factual signals by contrasting layers or model scales, e.g. DoLa [3], and compare against greedy and nucleus sampling.
3. Combine the best intervention with the abstention policy from Project B and report the net effect on confident-error rate.

Deliverables.

- An ablation table (grounding × decoding × abstention) reporting accuracy, faithfulness, and confident-error rate.
- A qualitative analysis of failure modes introduced by each intervention (e.g. over-copying from retrieved text, fluency degradation).
- A one-page recommendation on which combination best balances truthful abstention with usefulness.

Submission and assessment

Each project should be submitted as a reproducible repository (fixed seeds, environment file, and a `README` with exact commands), a 6–8 page report with figures, and a 10-minute recorded demo. Grading weights clarity of the experimental design, correctness of the calibration/statistical analysis, and depth of the mechanistic interpretation. Negative results are acceptable when accompanied by sound methodology.

References

[1] A. Azaria and T. Mitchell. The internal state of an LLM knows when it's lying. In *Findings of EMNLP*, 2023.

[2] C. Burns, H. Ye, D. Klein, and J. Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023.

[3] Y.-S. Chuang, Y. Xie, H. Luo, Y. Kim, J. Glass, and P. He. DoLa: Decoding by contrasting layers improves factuality in large language models. In *International Conference on Learning Representations (ICLR)*, 2024.

[4] S. Kadavath, T. Conerly, A. Askell, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

[5] A. T. Kalai, O. Nachum, S. S. Vempala, and E. Zhang. Why language models hallucinate. *arXiv preprint arXiv:2509.04664*, 2025.

[6] P. Lewis, E. Perez, A. Piktus, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[7] S. Lin, J. Hilton, and O. Evans. TruthfulQA: Measuring how models mimic human falsehoods. In *Proceedings of the Association for Computational Linguistics (ACL)*, 2022.

[8] P. Manakul, A. Liusie, and M. J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of EMNLP*, 2023.

[9] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald. On faithfulness and factuality in abstractive summarization. In *Proceedings of the Association for Computational Linguistics (ACL)*, 2020.

[10] OpenAI. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.